

Chenglin LIU

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[Linkedin](#) | [Google Scholar](#) | [Research Gate](#) | [Github](#)

EDUCATION EXPERIENCE

- Sichuan University of Science and Engineering** Sichuan, China 09/2015-06/2019
Bachelor of Information management and information system in School of Computer Science
- Southwest Jiaotong University** Sichuan, China 09/2021-06/2024
Master of Safety Engineering in School of Transportation and Logistics Supervisor: [Prof. Chaozhe Jiang](#)
- The Hong Kong University of Science and Technology** Guangzhou Campus, Guangzhou, China 02/2025-08/2026
The Hong Kong University of Science and Technology Clear Water Bay Campus, Hong Kong, China 08/2026-now
Doctor of Philosophy in Intelligent Transportation in Information and Transportation Thrust
- Supervisors: [Prof. Dengbo He](#) & [Pro. Hai Yang](#)

PUBLICATION

In Progress Manuscripts

1. Agent-Pilot: A Visual Hybrid Intelligence Closed-Loop Framework for Supervised Autonomous eVTOL Landing, **Communications in Transportation (IF:12.7, Q1)**. Liu, Chenglin, Du, Yuxin, Chen, Yufan, He, Dengbo*, Chen, Yingcong, and Yang, Hai. (Second revision). <https://github.com/Yuxin-Du-Lab/AgentPilot>
2. Training Ab Initio eVTOL Pilots for Regional Air Mobility Missions: A Comparison of Curricula, **Accident Analysis and Prevention (IF:8.7, Q1)** Chenglin Liu, Yuxin Du, Dengbo He*, Haoyuan Jiang, Chaozhe Jiang, Hai Yang. (Minor revision). https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5991728
3. The Relationships between Physiological and Eye-Tracking Responses and VR Motion Sickness, **INTERNATIONAL JOURNAL OF HUMAN-COMPUTER INTERACTION (IF:6.1, Q1)**. Chenglin Liu, Huawei Lin, Dengbo He*. (Under review)
4. High-Speed Rail Driver Stress Detection Using Multimodal Features of Electrocardiogram and Respiration Signals, **IEEE Transactions on Human-Machine Systems (IF:4.4, Q1)**. Kun Liu, Chenglin Liu*, Chen Tu, Yubo Jiao, Yunbiao Wang. (Second revision)
5. Progressive Two-Stage Training for Tiltrotor eVTOL Pilots: A Psychophysiological Mechanism Analysis Using Structural Equation Modeling, **Transportation Research Part C: Emerging Technologies (IF:8.5, Q1)**. Chenglin Liu, Yimin Wang, Zhenyu Wang, Dengbo He*, Hai Yang*. (Under review). https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7412882

Research papers

1. Liu, C.; Zhang, C.; Sun, L.; Liu, K.; Liu, H.; Zhu, W.; Jiang, C*. Detection of Pilot's Mental Workload Using a Wireless EEG Headset in Airfield Traffic Pattern Tasks. **Entropy** 2023, 25, 1035. <https://doi.org/10.3390/e25071035>. (IF:2.1, Q2)
2. Zhang, C.; Liu, C.; Liu, H.; Jiang, C.*; Fu, L.; Wen, C.; Cao, W. Incorporation of Pilot Factors into Risk Analysis of Civil Aviation Accidents from 2008 to 2020: A Data-Driven Bayesian Network Approach. **Aerospace** 2023, 10, 9. <https://doi.org/10.3390/aerospace10010009>. (IF:2.5, Q2)
3. Zhu, W.; Zhang, C.; Liu, C.; Yuan, J.; Li, X.; Wang, Y.; Jiang, C*. Recognition of pilot mental workload in the simulation operation of carrier-based aircraft using the portable EEG. **Proceedings of the 2023 3rd International Conference on Human Machine Interaction**, 2023.
4. Wang, Y.; Zhang, C.; Liu, C.; Liu, K.; Xu, F.; Yuan, J.; Jiang, C.*; Liu, C.; Cao, W. Analysis on pulse rate variability for pilot workload assessment based on wearable sensor. **Human Factors and Ergonomics in Manufacturing & Service Industries**, 34(6), 2024.
5. Wang, J.; Yan, S.; Yang, X.; He, Q.; Wang, A.; Liu, C.; Chen, C.; Wang, Z.; He, D*. From awareness to intent: Mitigating silent driving system failures through prospective situation awareness enhancing interfaces. **Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems**, 2026.

6. **C Liu**, C Zhang, Y Wang, F Xu, J He, C Jiang*, X Peng et al., "Identification of pilots' mental workload under different flight phases based on a portable EEG device," **2023 7th IEEE International Conference on Transportation Information and Safety (ICTIS)**, Xi'an, China, 2023, pp. 32-37, doi: 10.1109/ICTIS60134.2023.10243804.
7. **Chenglin Liu**, Wenjun Zhang, Dengbo He* et al., CTA-STGT: A Cortical Topology-Aware Spatiotemporal Graph Transformer for EEG-Based eVTOL Pilot Cognitive Workload Estimation, **18th International ACM Conference on Automotive User Interface '26 (AutoUI'26)** Gothenburg, Sweden. (Accepted)
8. Zhijie Yi, **Chenglin Liu**, Boyu Jin, Dengbo He*;Multimodal Physiological Fusion for Driver Fatigue Detection: Evidence from a 160-Hour Naturalistic Trucking Study, **19th The IEEE International Conference on Intelligent Transportation Systems (ITSC) 2026**, Naples, Italy. (Accepted)

ACADEMIC ACTIVITIES

Workshop Organizer, AutoUI '26 Gothenburg, Sweden 2026

- First organizer of "W14: Touchdown in the Urban Sky: Designing Human-Centered and Trustworthy Futures for eVTOLs and City Life" at the 2026 ACM International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutoUI). [Workshops Program – AutoUI '26](#)

Topic Coordinator, Frontiers in Computer Science 2026–2027

- Topic coordinator for the Research Topic "Human-AI Interaction in Agentic Transportation: Trust, Transparency, and Shared Decision-Making" in *Frontiers in Computer Science* (Section: Human-Media Interaction). [Frontiers | Human-AI Interaction in Agentic Transportation: Trust, Transparency, and Shared Decision-Making](#)

Peer Reviewer 2025–Present

- Served as an invited peer reviewer for multiple international journals, including *Accident Analysis and Prevention (AAP)*, *Reliability Engineering & System Safety*, *IEEE Transactions on Human-Machine Systems* etc.

Oral presentation on the 2023 7th International Conference on Transportation Information and Safety (ICTIS), Xi'an, China, 2023.

WORK EXPERIENCE

03/2023-04/2023 Intern at Sichuan Emergency Fire Protection Research Institute.	Chengdu, China
06/2024-10/2024 Technician at China Railway Chengdu Group Co., Ltd	Chengdu, China
10/2024-02/2025 Research Assistant at The Hong Kong University of Science and Technology(Guangzhou), Guangzhou, China	

HONOR

The China Graduate National Scholarship	2023
The First-Prize Scholarship, twice (top 10%, SWJTU)	2022
19th May Day Mathematical Contest in Modeling <i>2nd Prize</i>	2022
19th 'Huawei cup' China Post-graduate Mathematical Contest in Modeling <i>2nd Prize</i>	2023
13th National College Student Market Survey and Analysis <i>2nd Prize</i>	2023